ECON2010H

INTERMEDIATE MACROECONOMIC THEORY I

WEEK 04/05: A ONE-PERIOD MACROECONOMIC MODEL IN A CLOSED ECONOMY

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READING: TEXTBOOK CHAPTER 5

OUTLINE

1. Introduction
2. Government
3. The model and competitive equilibrium
4. Pareto optimality
5. Effects of change government spending
6. Effects of change in total factor productivity
7. Effects of change in tax rate
8. Keynesian Sticky wage and prices

0. INTRODUCTION

- In Chapter 4, we studied the optimal choices of the representative consumer and the representative firm
- The model economy is perfectly competitive, consumer and firm take prices as given
- Now, we build the microeconomic behavior of consumer and firm into the first macroeconomic model
- We will define and characterize the **competitive equilibrium** for a closed economy
- The model is then used to examine the effects of governmental policy.

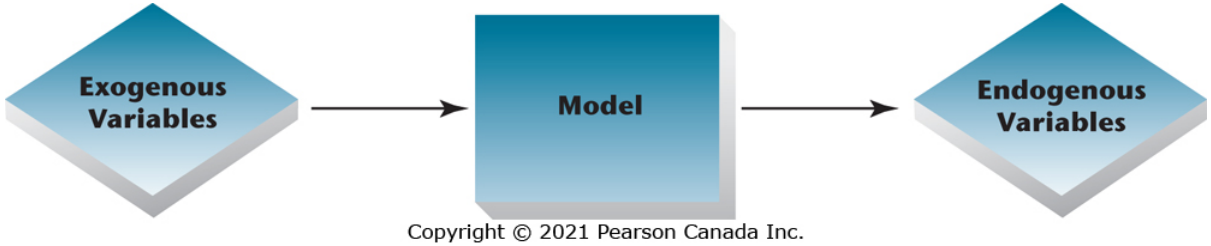
1. GOVERNMENT

- Government provides public goods, e.g, national defense, that are difficult or impossible for the private sector to provide
- Let G be the real government spending on providing public goods
- G is simply given, it is an **exogenous variable** as we do not address what level of G should be
    - G does not depend on what happens in the rest of the economy

- Government budget constraint is $G = T$
    - Government does not borrow or save
- This is the simplest way to study effects of fiscal policy in a one-period model
- **Fiscal policy** refers to the government's choices over its expenditure, taxes, transfers, and borrowing

2. THE MODEL AND THE COMPETITIVE EQUILIBRIUM

- A macroeconomic model takes the exogenous variables, and determines the values for endogenous variables



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2-1. A ONE-PERIOD MACROECONOMIC MODEL

- The model economy has three players:
    - The representative consumer
    - The representative firm
    - Government
- All markets are perfectly competitive, consumer and firm take prices as given
    - Recall, the representative consumer is a stand-in for all consumers
    - And, the CRS technology implies that the representative firm behaves the same way as if there are many firms

- In the macroeconomic model we are building, the exogenous variables are G, z, K
    - G, government spending
    - z, total factor productivity
    - K, capital stock
- And, endogenous variables are C, N^s, N^d, Y, T, w
    - C, quantity of consumption
    - N^s, quantity of labor supplied
    - N^d, quantity of labor demanded
    - Y, quantity of output (real GDP)
    - T, lump-sum tax
    - w, real wage rate

2-2. COMPETITIVE EQUILIBRIUM

- A **competitive equilibrium** is a state in which, given some market prices, quantity demanded equals quantity supplied in all markets
- In the model, consumer and firm take prices as given, the economy is perfectly competitive
- When quantities demanded and supplied equal, **markets clear**

- A competitive equilibrium is a set of endogenous quantities, C, N^s, N^d, T, Y, and w, such that, given the exogenous variables G, z, and K, the following are satisfied:
    - The representative consumer optimizes, utility is maximized
    - The representative firm optimizes, profit is maximized
    - The labor market clears, $N^s(w) = N^d(w)$
    - The government budget constraint is satisfied, $G = T$
- Note: all endogenous quantities should have a superscript o or asterisk to indicate that these quantities are values in competitive equilibrium, I did not, to make them less tedious

AN IMPORTANT PROPERTY OF A COMPETITIVE EQUILIBRIUM

- A property of a competitive equilibrium is: $Y = C + G$
    - It is the income-expenditure identity we saw in measurement
    - Our model is static, there is no investment
    - Our model is for a closed economy, $NX = 0$
- To see this, recall consumer's budget constraint, $C = wN^s + \pi - T$. Dividend (profit) income $\pi = Y - wN^d$
- Substitute for π in consumer's budget constraint, then $C = wN^s + Y - wN^d - T$
- In equilibrium, $N^s = N^d$ and $T = G$. Therefore, $Y = C + G$
- The income-expenditure identity is not imposed, but a result of the model equilibrium

REMARKS ON SOLVING THE COMPETITIVE EQUILIBRIUM

- In competitive equilibrium, prices are determined to clear the markets
- In our model, real wage is determined to clear the labor market, i.e., $N^s = N^d$
- In general, solving for prices in equilibrium requires numerical methods, closed-form solution is not possible
- Our model is simple enough to use diagrams to analyze equilibrium

2-2. DIAGRAM ANALYSIS OF COMPETITIVE EQUILIBRIUM

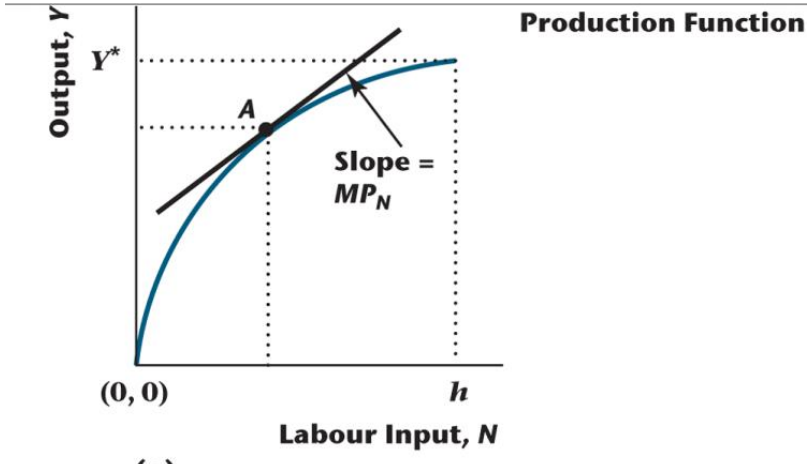
- In order to analyze competitive equilibrium using diagrams, we need to integrate the behavior of consumer and of firm on the same diagram
- We proceed in two steps
    - Production possibilities frontier
    - Competitive equilibrium analyzed on diagram

PRODUCTION POSSIBILITIES FRONTIER

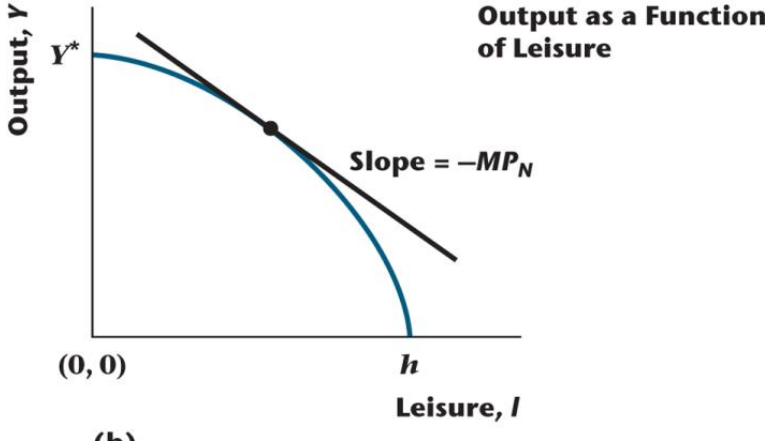
- Production possibilities frontier shows the maximum amounts of outputs can be produced when resources are fully utilized in production.
- Leisure l is an output in the economy. For each l, the labor input is $N^d = h - l$ (labor is fully used).
- Re-write production function as $Y = zF(K, h - l)$, and noting $Y = C + G$, then

$$C = zF(K, h - l) - G \tag{1}$$

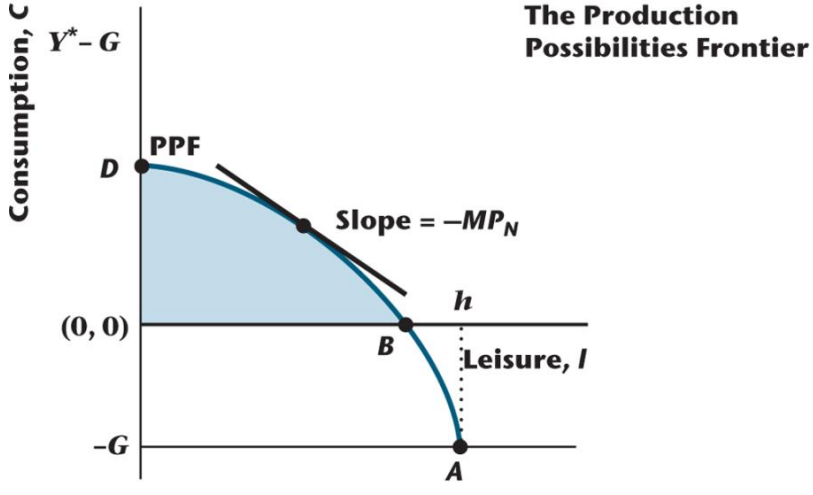
- This equation is the **production possibilities frontier (PPF)**: the relationship between C and l through the production function and equilibrium property $Y = C + G$
- PPF describes the technological possibilities of producing the consumption good and leisure
- Production function describes the technology transforming inputs into output. PPF describes the same technology but in terms of how much consumer goods (consumption and leisure) can be provided


Production Function
Output, Y
Y*
A
Slope = MP_N
(0, 0)
h
Labour Input, N

(a)


Output as a Function of Leisure
Output, Y
Y*
Slope = –MP_N
(0, 0)
h
Leisure, l

(b)


The Production Possibilities Frontier
Consumption, C
Y* – G
PPF
D
Slope = –MP_N
h
(0, 0)
B
Leisure, l
–G
A

(c)

- At any point on the PPF: output is fully used ($Y = C + G$), and labor input is fully used ($N^d = N^s$)
    - Note: Labor input being fully used means that the PPF shows maximum outputs if resources are fully used.
- on the PPF, the **marginal rate of transformation (MRT)** is the rate at which one good can be converted technologically into another
- In our model, the MRT is the rate at which leisure is converted into consumption goods through work, we write it as $MRT_{l,C}$

$$MRT_{l,C} = MP_N = \text{the negative of slope of the PPF} \tag{2}$$

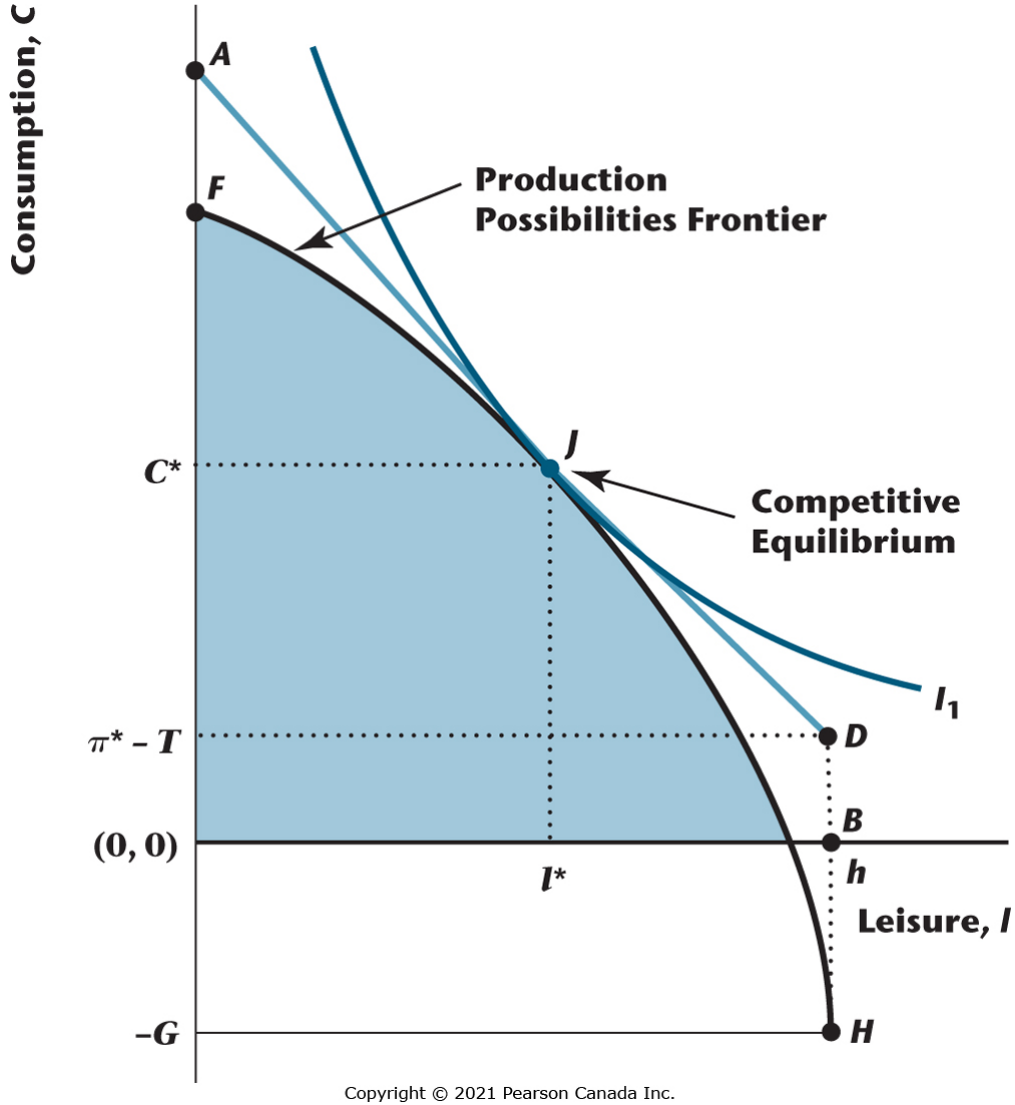
- Recall, the optimality condition is $MP_N = w$ for the representative firm
- Thus, on the PPF, the optimal bundle (l, C) *supplied* satisfies $MRT_{l,C} = w$.
- Note: the MRT of leisure into consumption is a positive ratio

INTEGRATING PPF WITH THE CONSUMER'S PROBLEM

- Recall, the optimal consumer choice (l, C) satisfies $MRS_{l,C} = w$, which says that choices are optimal at the point on indifference curve where the negative slope of tangent line equals w
- Putting together PPF and consumer's problem, leading to the graph next page
- A competitive equilibrium is achieved when the optimal bundle (l, C) supplied coincides the optimal bundle chosen by the consumer
- Mathematically, the economy is in competitive equilibrium when

$$MRT_{l,C} = MRS_{l,C} = w \tag{3}$$

- It says, the marginal rate of substitution of leisure for consumption is equal to the marginal rate of transformation of leisure into consumption



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3. OPTIMALITY

- In competitive equilibrium, the optimal quantities supplied and the optimal quantities demanded equal
- In equilibrium, both profit and utility are maximized, through the price (real wage).
- Is the competitive equilibrium socially optimal? Is there a better allocation than that achieved in competitive equilibrium?
- We introduce the Pareto optimality, a concept of social **efficiency**.

3-1. PARETO OPTIMALITY

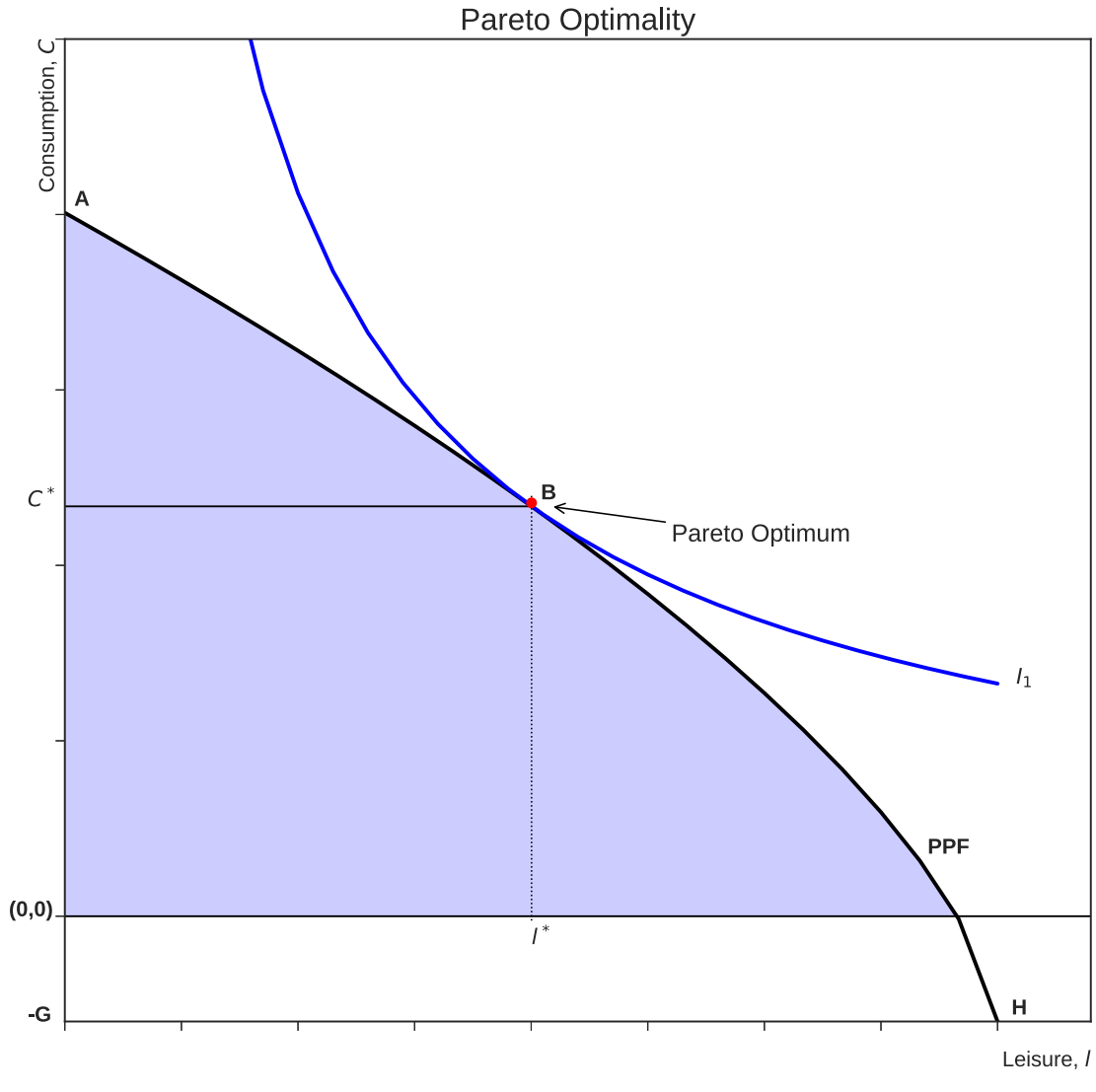
- A combination of goods (allocation of goods) is **Pareto optimal** if there is no way to rearrange production or to reallocate goods so that someone is made better off without making someone else worse off

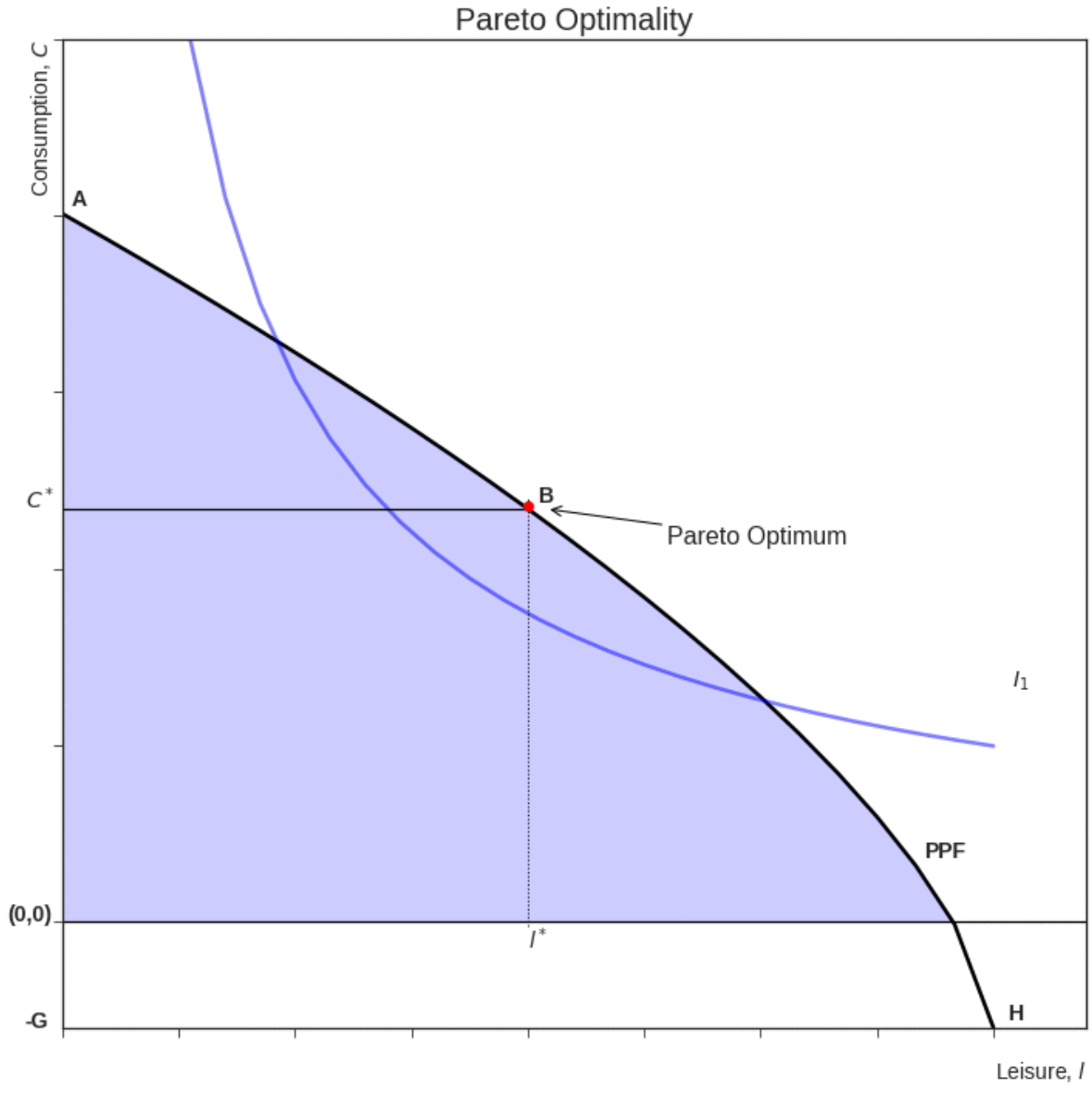
- In other words, an allocation is Pareto optimal if any deviation from the allocation makes someone in the economy worse off

- To find the Pareto optimum, we bring in a fictitious social planner, a benevolent dictator, whose objective to maximize the consumer's utility,

    - but not through markets
- The social planner's optimization problem is to make the representative consumer as well off as possible by choosing a bundle (l, C), given the production technology

- In the Pareto optimum, the allocation (l, C) must be technologically possible (on the PPF), it must also make indifference curve as far as possible away from the origin


Pareto Optimality
Consumption, C
A
B
C*
Pareto Optimum
I1
PPF
(0,0)
l*
-G
H
Leisure, l


Pareto Optimality
Consumption, C
A
C*
B
Pareto Optimum
I1
PPF
(0,0)
l*
-G
H
Leisure, l

- At point (l^*, C^*), utility is maximized and technologically possible
    - Any bundle greater than (l^*, C^*) is technologically impossible
    - Any bundle smaller than (l^*, C^*) is in the shaded area, technologically feasible, but utility can be further increased
- Two questions concerning Pareto optimality and competitive equilibrium
    - Whether a competitive equilibrium is Pareto optimal
    - Whether Pareto optimality can be achieved through free markets
- Two theorems provide answers to the two questions

3-2. FUNDAMENTAL THEOREM OF WELFARE ECONOMICS

> *The first fundamental theorem of welfare economics states that, under certain conditions, a competitive equilibrium is Pareto-optimal*
>
> *The second fundamental theorem of welfare economics states that, under certain conditions, a Pareto optimum can be implemented by a competitive equilibrium*

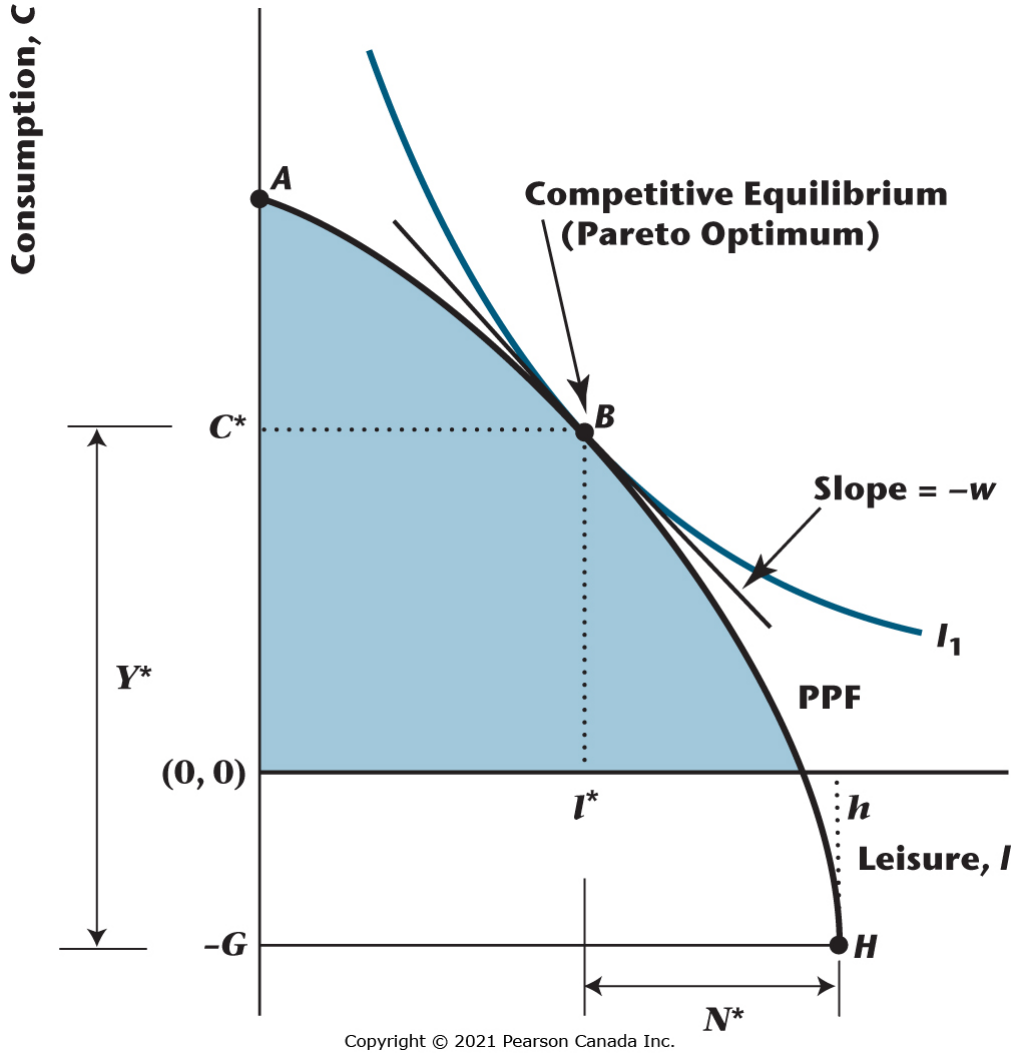
- Previous diagrams show that competitive equilibrium of the model we have constructed is indeed Pareto-optimal
- The first fundamental theorem suggests that, under certain conditions, free markets achieve Pareto optimality
- By the second fundamental theorem, the benevolent dictator can rely on free market to implement his/her plan

- A competitive equilibrium fails to be Pareto optimal if any of the following occurs:
    - the economy has externality
    - there are distorting taxes
    - market competition is imperfect, i.e., firms are not price takers
- There are other conditions under which the first fundamental theorem fails to hold, we skip them
- **Externality** is any activity for which an individual firm or consumer does not take account of all associated costs and benefits, e.g., air pollution
- Competitive equilibrium in the presence of externality loses efficiency due to **market failures**

- **Distortionary taxes** are those that depend on actions of individuals being taxed
- An example, proportional labor income tax, $tw(h-l)$, where t is tax rate. The consumer's effective labor income is $(1-t)w(h-l)$
- With a proportional labor income tax, the consumer's optimality condition becomes $MRS_{l,C} = w(1-t)$
- The firm's optimality condition remains unchanged, $MRT_{l,C} = MP_n = w$
- Then, $MRS_{l,C} < MRT_{l,C}$, competitive equilibrium is not Pareto optimal

- **Market power**. If firms are have market power, they can influence or determine the price
    - $MP_n = w$ is no longer optimal for the firm
- Competitive equilibrium is then no longer Pareto optimal
- Very recent studies show that, increased market power in both output markets and in labor market in the U.S. contributed to increased inequality in income and in wealth
- Will visit this issue when we study economic growth
- Let me know if you would like a discussion on this

- Finally, the second fundamental theorem of welfare economics, illustrated



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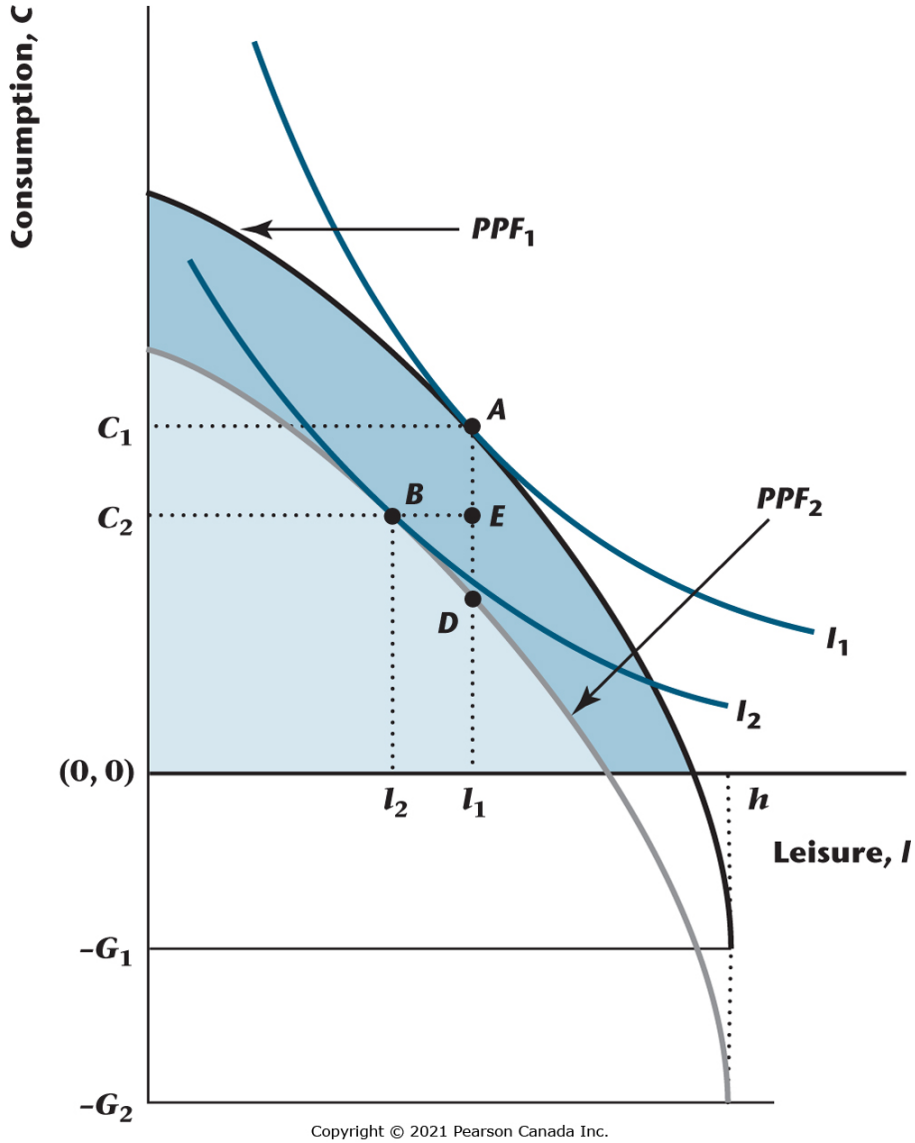
TRANSITION SLIDE

- In the rest of this chapter, we carry out thought experiments using the general equilibrium model
- In each experiment, we compare equilibrium before and after a change in an exogenous variable
- We do not examine the process of adjusting from the initial equilibrium to the new equilibrium following a change in exogenous variables
- One should note that, the equilibrium model studied here is a benchmark, not universally applicable
- The model is consistent with data in some aspects, but not in other
- The principles and methods applied to construct the model are nevertheless useful and widely accepted

4. EFFECTS OF A CHANGE IN GOVERNMENT SPENDING

- The first experiment: the effects of an increase in government spending
- Such increase shifts PPF downward, leading to lower and consumption and less leisure
- Quantity of output rises because consumer works more
- $\Delta C = \Delta Y - \Delta G$. $\Delta C < 0$if $\Delta G > 0$
    - private consumption is **crowded out** by government spending
    - but consumption is not completely crowded out, since $\Delta Y > 0$ when $\Delta G > 0$
- Real Wage must fall, why?



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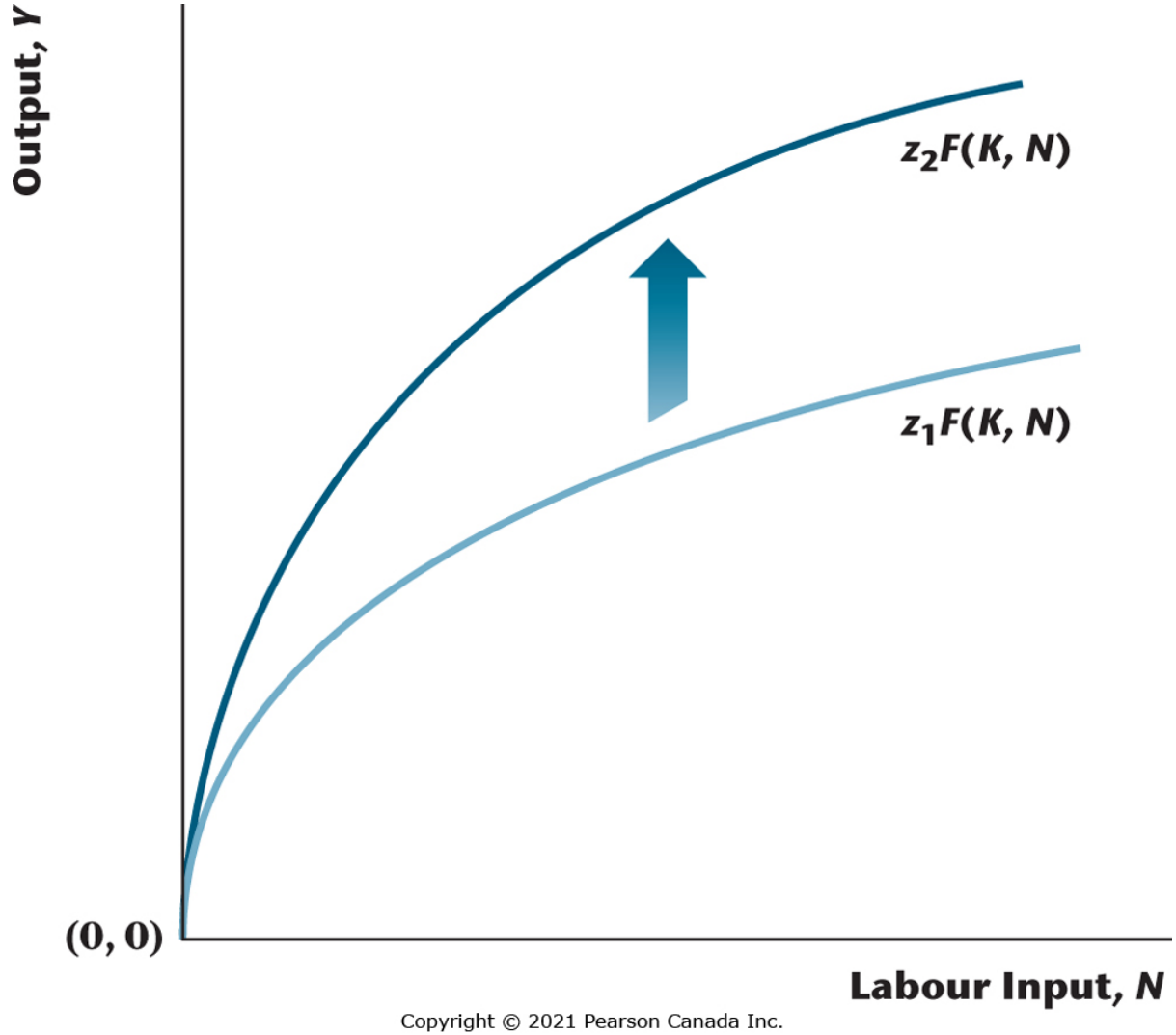
- Question: Are fluctuations in government spending a cause of business cycles?
    - Yes, predicted by the model
    - Unlikely, because the model is inconsistent with facts of the business cycle
- The model prediction: If $\Delta G > 0$, $\Delta Y > 0$, $\Delta < 0$, $\Delta N > 0$, $\Delta w < 0$
- In other words, model predicts that
    - employment is procyclical (output increases, employment increases)
    - consumption and real wage are countercyclical

- Model prediction is consistent with facts regarding employment
- But the model prediction is inconsistent with other facts of the business cycle, facts are
    - consumption is procyclical, real wage is procyclical
- Therefore, the model prediction that government spending causes business cycles may not hold in the data

5. EFFECTS OF A CHANGE IN TFP

- The effects of an increase in z, total factor productivity (TFP)
- Many causes of improvements in TFP, all are outside our model
- An increase in TFP shifts upward both production function and marginal product of labor
- MP_N increases, the PPF becomes steeper
- Real wage must increase too, since in equilibrium $MP_n = w$



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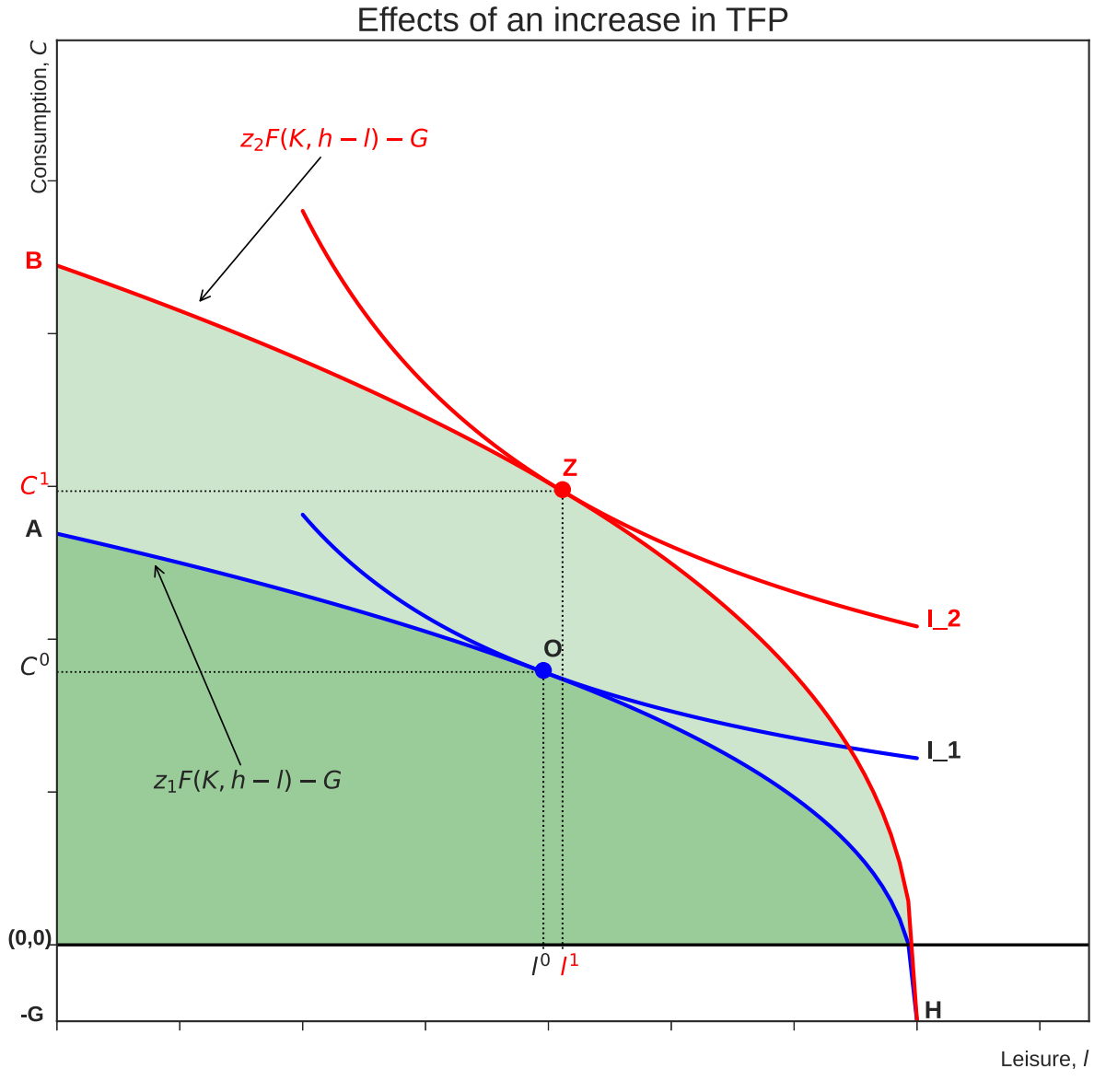
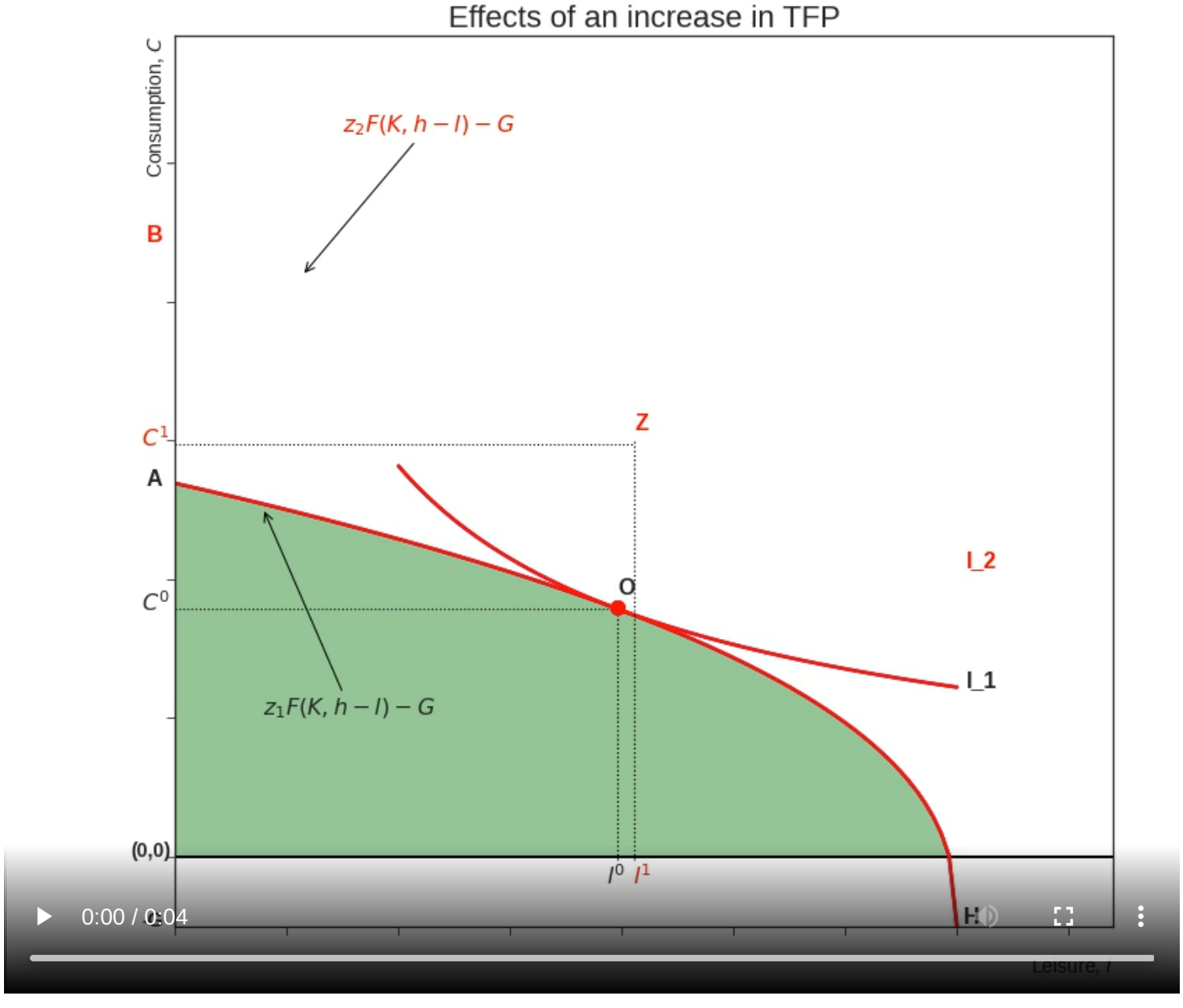
- The PPF moves from AOH to BZH, equilibrium consumption increases from C^0 to C^1
- Equilibrium output then must increases since $Y = C + G$ and G remains unchanged
- Does leisure increase, decreases, or remain unchanged?
- Quantity of leisure shown in graphs below increases from l^0 to l^1
- In general, whether the quantity of leisure increases or decreases depends on relative size of the substitution effect and the income effect of an increase in TFP

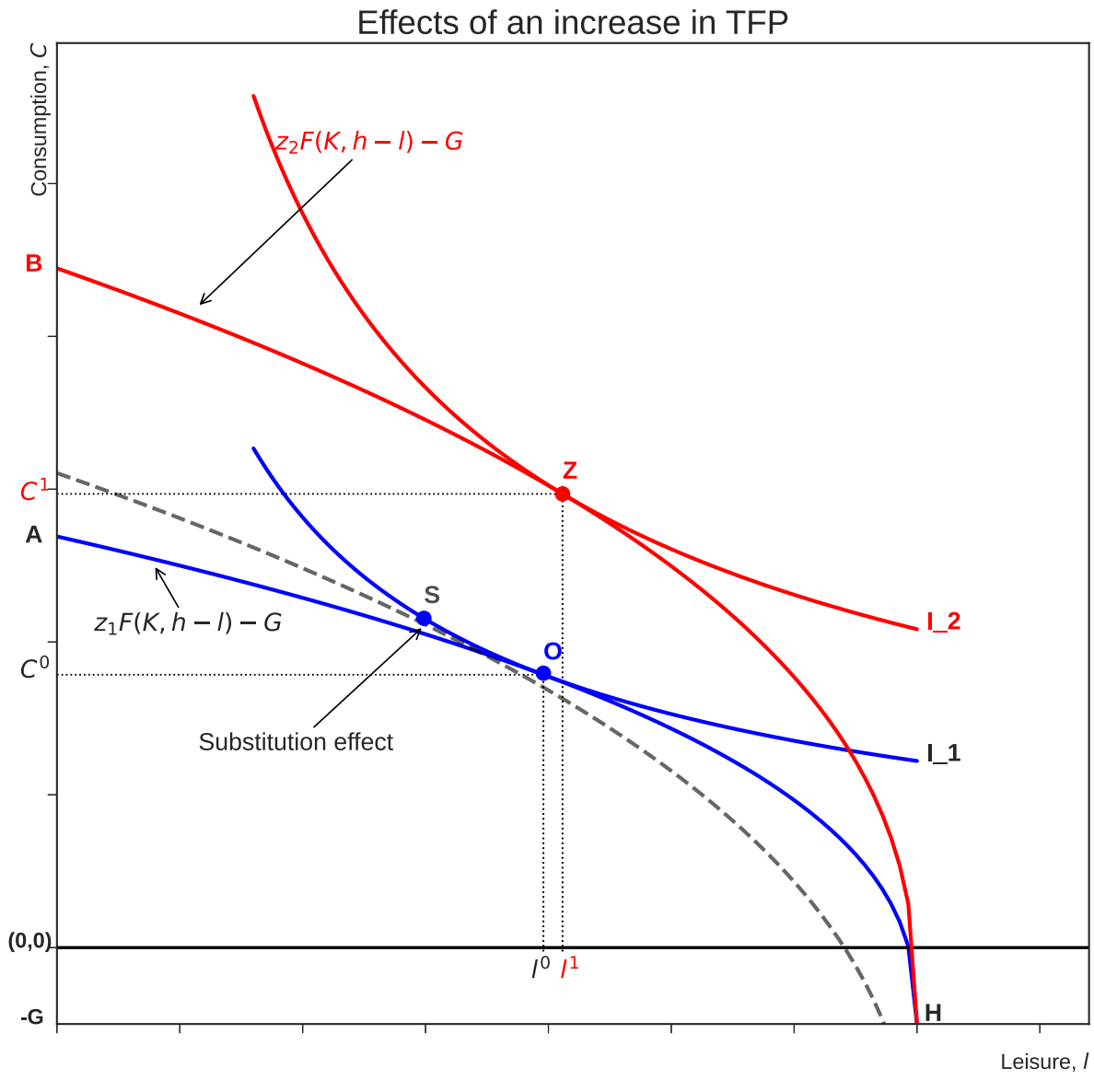

Effects of an increase in TFP
Consumption, C
$z_2F(K, h-l) - G$
B
Z
C^1
A
I_2
O
C^0
I_1
$z_1F(K, h-l) - G$
(0,0)
l^0
l^1
-G
H
Leisure, l

EFFECTS OF AN INCREASE IN TFP



THE SUBSTITUTION EFFECT

- Recall, the substitution effect is the change in the optimal bundle purely resulted from a change in the relative price (real wage)
- The substitution effect is calculated by taking away some income after the TFP increases such that the consumer's utility remains the same as before the TFP increases
- The PPF for the substitution effect is $C = z_2F(K, h - l) - G - C_0$, where C_0 is the amount of income taken away
- Point S in graph of next page is the resulted substitution effect
- An increase in TFP results in a higher real wage, meaning the relative price of leisure increases
- The consumer then substitutes more consumption for leisure: higher consumption, lower leisure


Effects of an increase in TFP
Consumption, C
$z_2F(K, h-l)-G$
B
Z
C^1
A
S
$z_1F(K, h-l)-G$
O
C^0
Substitution effect
I_2
I_1
(0,0)
l^0
l^1
-G
H
Leisure, l

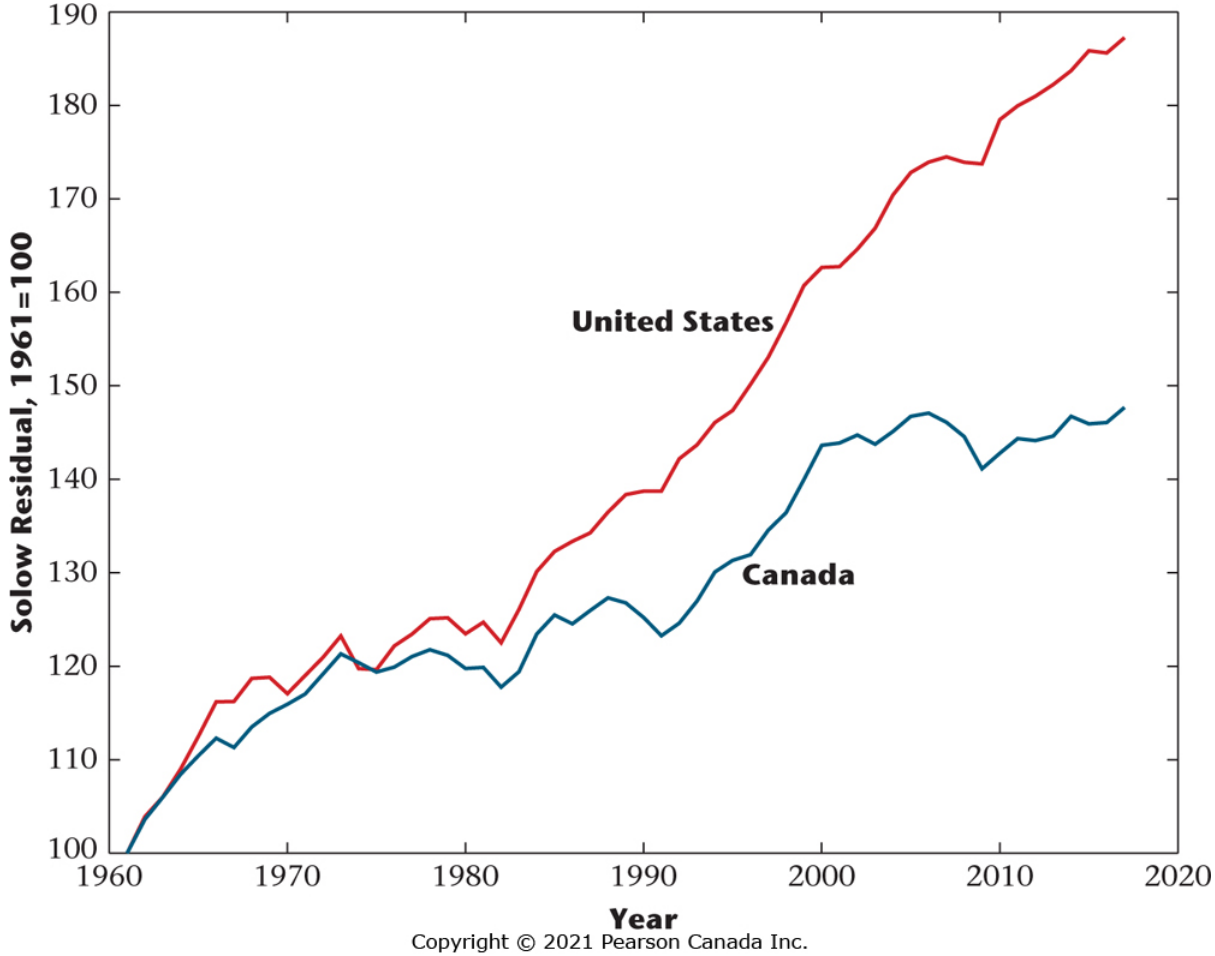
THE INCOME EFFECT

- On top of the substitution effect (point S in the above graph), we add the income take away, C_0, back to the consumer
- the PPF and the consumer's budget constraint shift outward
- Starting from the substitution effect, an increase in income by C_0 raises consumption and leisure
- Income effect: higher consumption, higher leisure

THE TOTAL EFFECT OF AN INCREASE IN TFP

- Higher consumption
- Ambiguous change in leisure: depending on which effect dominates (graph shows a case of dominant income effect)

- TFP and GDP



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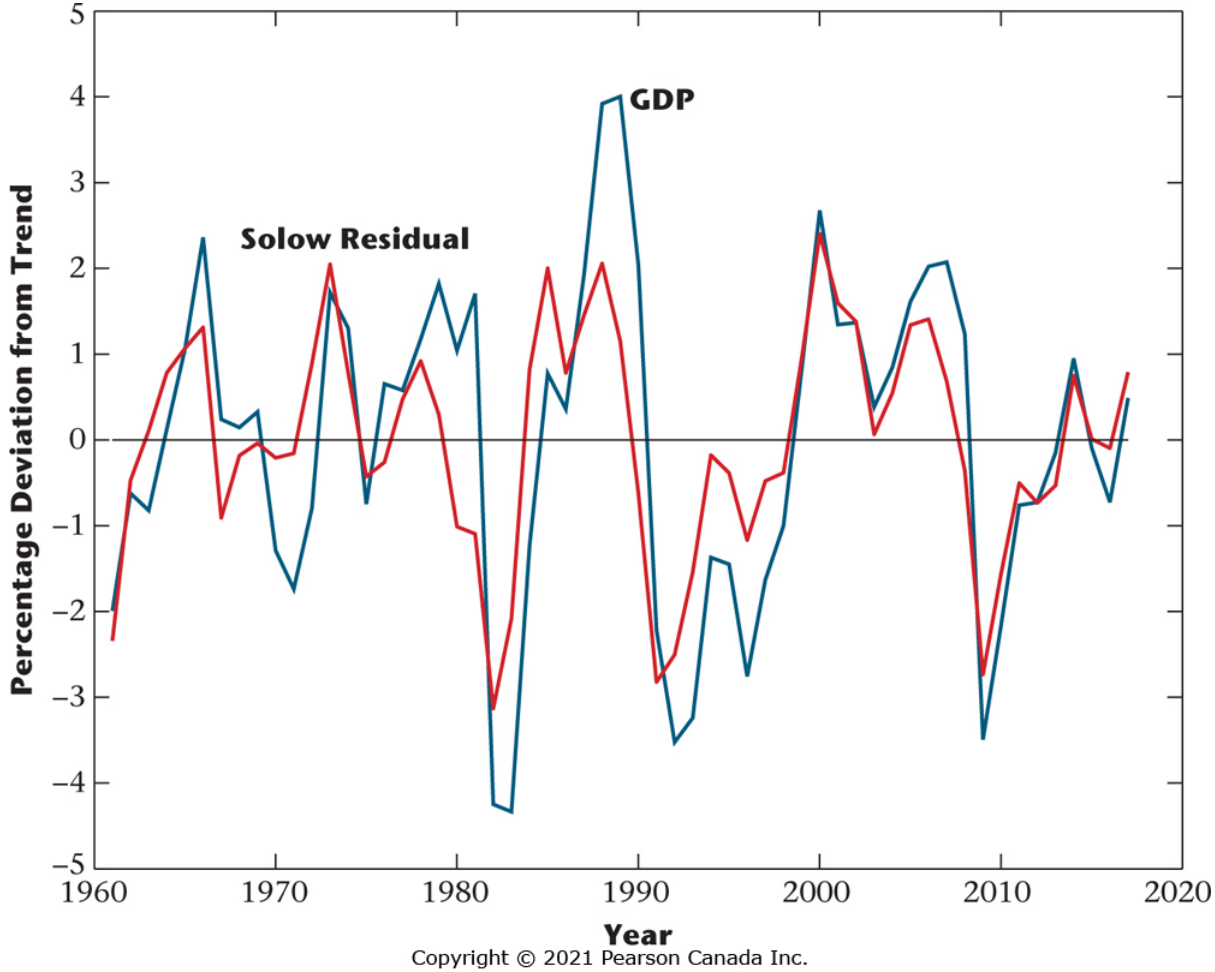

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6. EFFECTS OF A CHANGE IN DISTORTIONARY TAX ON LABOR INCOME, AND THE LAFFER CURVE

- We consider a variation of the model, to take into account distortionary labor income tax
- In the presence of distortionary taxes, competitive equilibrium is not Pareto optimal
- We cannot use Pareto optimality to study competitive equilibrium.
- The model shows that, theoretically, it is possible to increase tax revenue by lowering income tax rate
- The non-monotonic relationship between tax rate and tax revenue is known as the Laffer curve

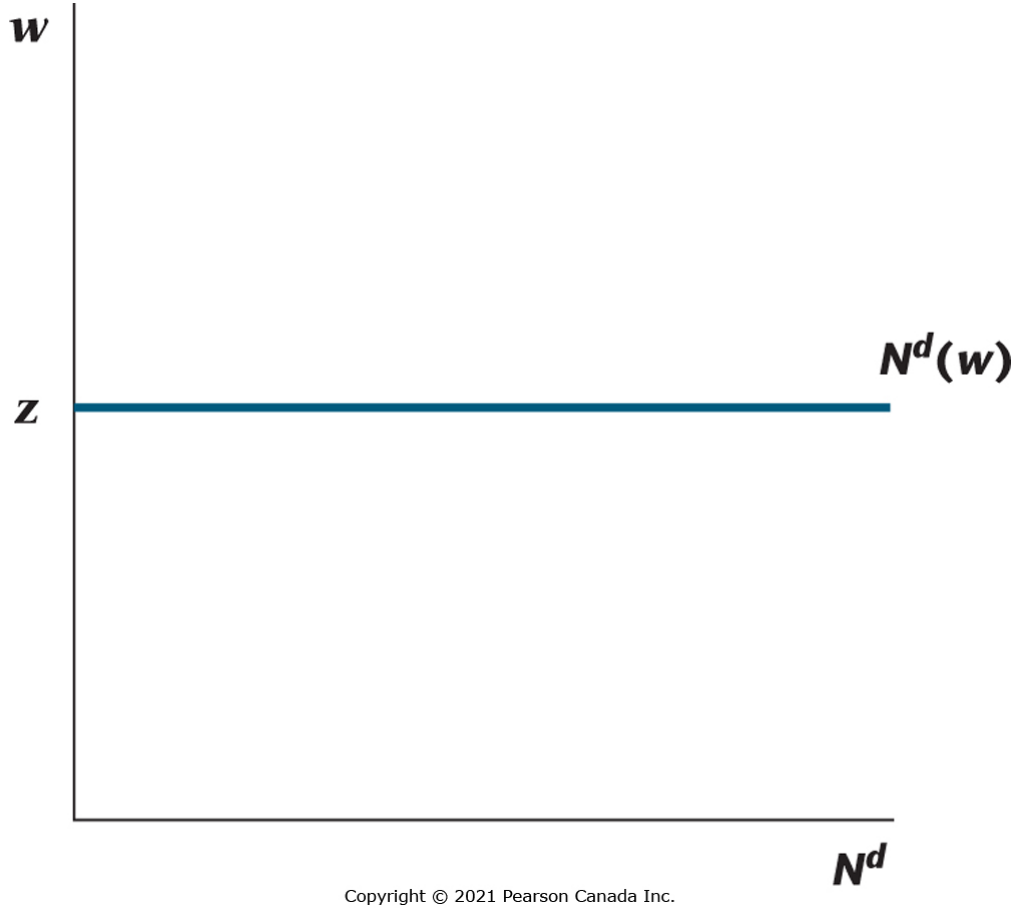
6-1. A SIMPLIFIED ONE-PERIOD MODEL WITH PROPORTIONAL INCOME TAXATION

- To keep the analysis simple and transparent, we simplify the model
- **Production by the representative firm**: assume that output is produced only with the labor input, and

$$Y = zN^d$$

  - Y is aggregate output, N^d is labor input, z is total factor productivity
  - The production technology exhibits CRS
- **Profit** for the representative firm is $\pi = Y - wN^d = (z - w)N^d$.
- Profit maximization leads to the following demand for labor
  - If $z > w$, N^d is infinity
  - If $z < w$, $N^d = 0$, otherwise profits are negative.
  - If $z = w$, N^d is infinitely elastic, since the firm is indifferent in how much to hire.

- Labor demand curve is infinitely elastic at $w = z$



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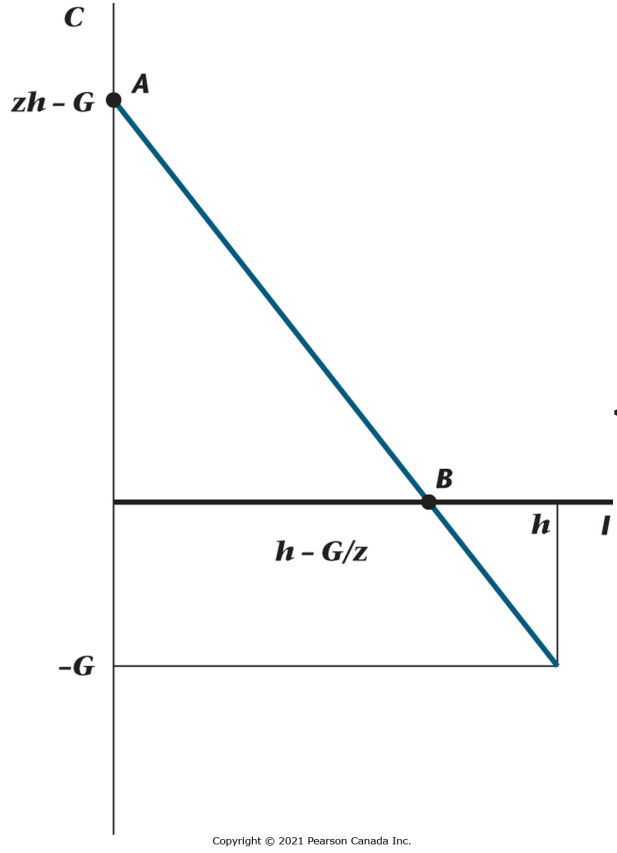
- The PPF is drawn when output and labor input are fully used: $C + G = Y$ and $N^d = h - l$
- Substituting $h - l$ for N^d in production function, **the PPF** is given by $C = z(h - l) - G$



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- **Consumer**'s budget constraint is given by

$$C = w(1-t)(h-l) + \pi$$

  - Tax rate is t, and $w(1-t)$ is effective real wage rate
- The optimality condition for the consumer's problem is $MRS_{l,C} = w(1-t)$
- Given the labor demand, real wage in equilibrium must equal TFP, or $w = z$, regardless of the consumer's behavior
- Zero profit, $\pi = 0$; The consumer's budget constraint becomes $C = z(1-t)(h-l)$
- Note, in this setup, proportional tax rate does not distort real wage rate, a special case

THE COMPETITIVE EQUILIBRIUM CONSISTS OF THE FOLLOWING

- Given the real wage rate, the representative consumer maximizes utility by choosing leisure l and consumption C subject to budget constraint
- Given the real wage rate, the representative firm maximizes profits
- Real wage rate satisfies $w = z$
- Total resource (or market clearing): $C + G = Y$, that is, $C = z(h - l) - G$

- **Government** budget constraint is balanced in competitive equilibrium
- How? By the output market clearing and the consumer's budget constraint
- Aggregate output $z(h-l) = C + G$, budget constraint $C = w(1-t)(h-l)$. Noting: $w = z$
- The two equations give $G = zt(h-l)$, government spending equals tax revenue
- Then, is G or t exogenously given? We assume that G is exogenous
- It is not crucial. We can think of one of the follow situations:
    - Tax rate t is exogenously given, tax revenue is used for spending G, the value of G in equilibrium depends on how much tax revenue is collected
    - G is exogenously given, government adjusts tax rate t, such that $G = zt(h-l)$ indeed holds in equilibrium
- In the analysis below, we take G as given (exogenous), thus the model determines t

SUMMARY OF THE MODEL

- The **firm's profit**: $\pi = zN^d - wN^d = (z - w)N^d$
- The only meaningful case is $w = z$, under which the demand for labor is horizontal
- Profit maximization leads to zero profit, $\pi = 0$
- The **consumer's utility** is maximized if $\mathrm{MRS}_{l,C} = (1 - t)w$ and $C = (1 - t)w(h - l) + \pi$
- **Government** determines tax rate t, and $G = tw(h - l)$
- **Competitive equilibrium** (CE) is a set of quantities (not variables):
  - $w^* = z$, $N^* = h - l^*$
  - $C^* = (1 - t)w^*(h - l^*)$, $\mathrm{MRS}_{l^*,C^*} = (1 - t)w^*$;
  - $G = t^*w^*(h - l^*)$
- CE is on the PPF: $C^* = z(h - l^*) - G$, then $C^* = w^*(h - l^*) - tw^*(h - l^*) = (1 - t)w^*(h - l^*)$

- Note: letters with asterisks are quantities in CE, not variables. Sometimes, we omit the asterisks

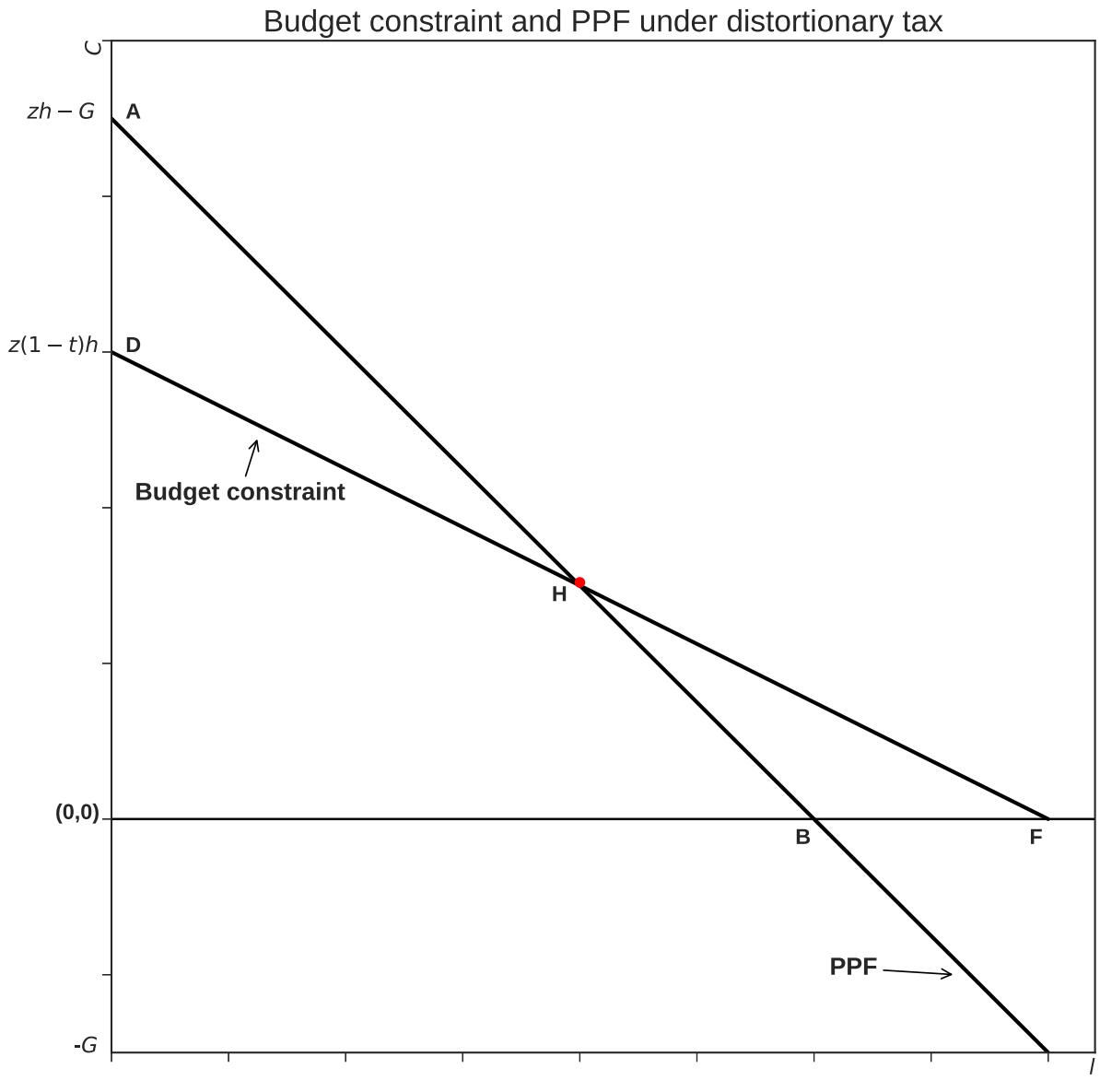
PARETO OPTIMALITY IN EQUATIONS

- G is set at the same level as in competitive equilibrium
- PPF $C = z(h - l) - G$ is a straight line, thus MRT is constant, equal to z
- **Pareto optimality**: $C^o = z(h - l^o) - G$ and $\mathrm{MRS}_{l^o,C^o} = \mathrm{MRT}_{l^o,C^o}$ where $\mathrm{MRT}_{l,C} = z$
    - Letters with superscript o are quantities that are Pareto optimal
- Compare CE and Pareto optimality: $C^o > C^*$; $l^o < l^*$; $Y^o > Y^*$
- Why, the effective price of leisure in CE is $z(1 - t)$ which is less than z
- Proportional tax is a disincentive to work for more hours

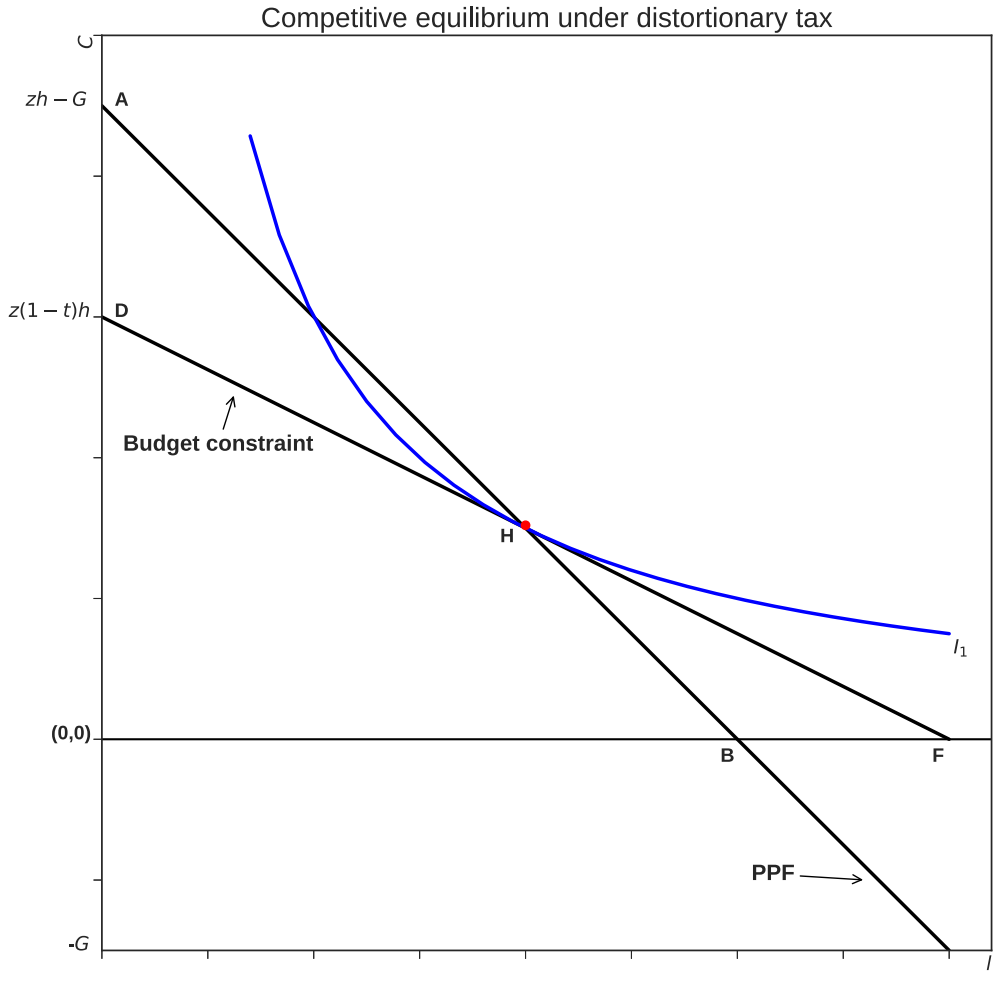
6-2. COMPETITIVE EQUILIBRIUM AND PARETO OPTIMALITY

- A competitive equilibrium (CE) in the presence of distortionary tax is not Pareto optimal (PO)
- The difference between CE and PO is seen in the difference of the PPF and the consumer's budget constraint
- PPF is again $C = z(h - l) - G$. Note that G is given at the same level as in CE, social planner does not determine it
- The consumer's budget constraint is $C = z(1 - t)(h - l)$
- **Claim**: The PPF is steeper than the consumer's budget constraint
    - Slope of the PPF is $-z$; slope of the budget constraint is $-z(1 - t)$
    - Note, $0 < t < 1$

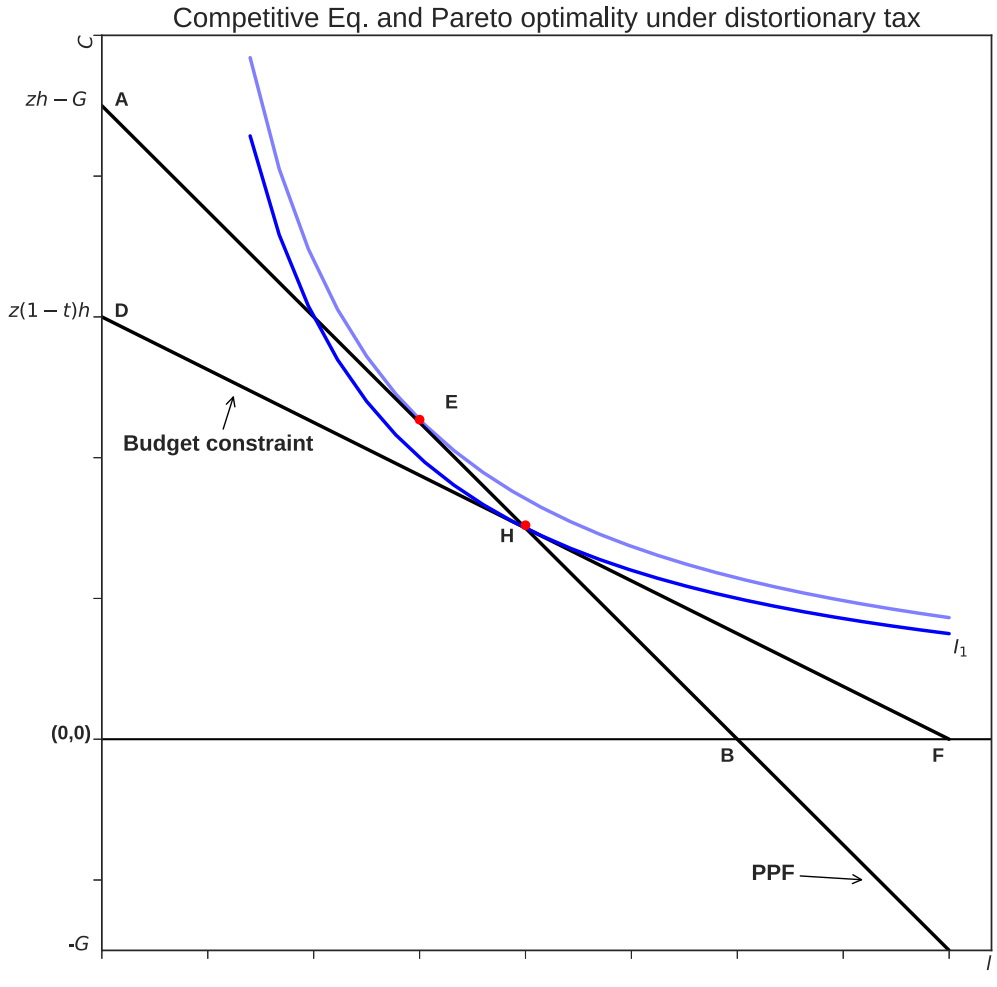
Budget constraint and the PPF



- The competitive equilibrium is point H on next diagram
- It is on the consumer's budget constraint line DHF, and on the indifference curve I_1.



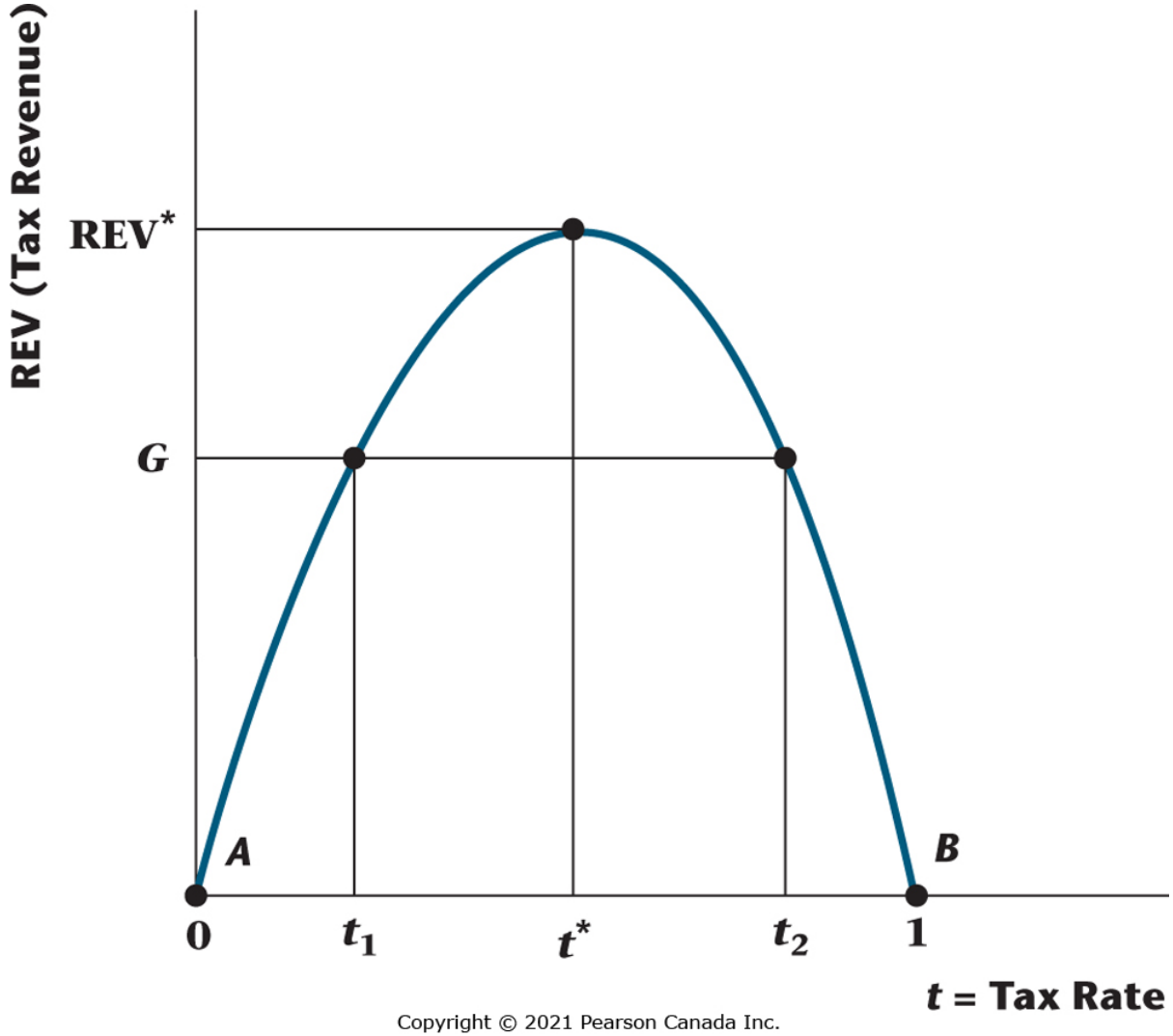
- Note, that point H must be on the PPF, since the CE is technologically feasible
- Pareto optimality is the allocation determined by the benevolent social planner
    - No price system is used
    - G is at the same level as that in competitive equilibrium
- The allocation (l, C) that is Pareto optimal is the tangent point of the indifference curve and the PPF
    - It is point E in the diagram of the next page
- Socially efficient allocation (point E): higher output and consumption, lower leisure, higher utility
- Competitive equilibrium allocation (point H): lower output and consumption, more leisure, lower utility
- Intuition: proportional tax reduces the incentive to work


Competitive Eq. and Pareto optimality under distortionary tax
C
zh − G
A
z(1 − t)h
D
E
Budget constraint
H
I1
(0,0)
B
F
PPF
-G
l

6-3. INCOME TAX REVENUE AND THE LAFFER CURVE

- We now re-visit government spending G and tax rate t
- A basic question: to meet a demand for higher spending G, should government raise tax rate?
- Raising tax rate increases revenue for each unit of labor income
- But, higher tax rate makes the consumer work less, reducing labor income
- On net, higher tax rate may or may not lead to higher tax revenue

- Consider the model we studied, tax revenue is $tz[h - l(t)]$
- Before-tax labor income, $z[h - l(t)]$, is the **tax base**
- $l(t)$ is the equilibrium leisure which depends on tax rate
    - When tax rate is higher, $l(t)$ increases, tax base becomes smaller.
- Thus, tax revenue may or may not increase
- The **Laffer curve** shows that there is a level of tax rate at which tax revenue is the highest



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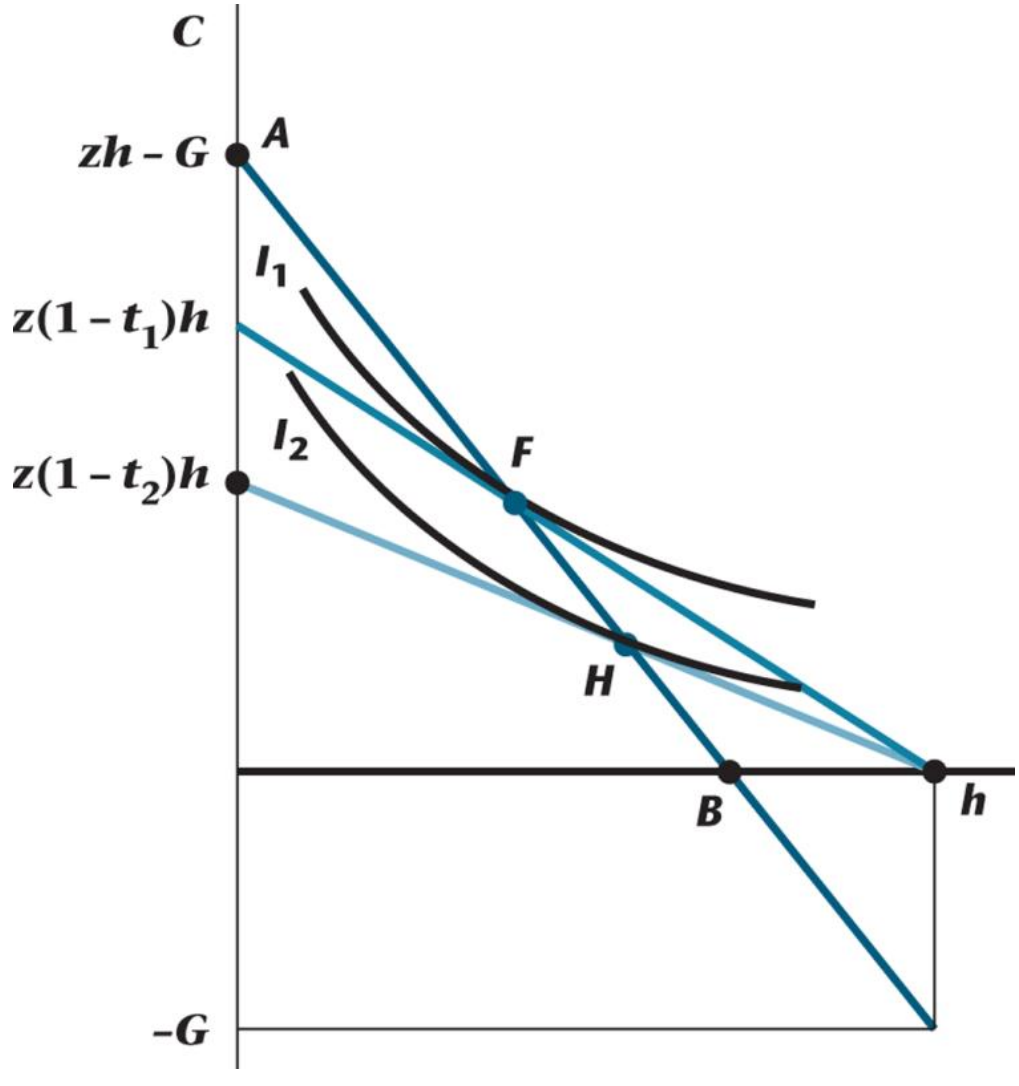
- Given a level of government spending G, there may be two tax rates to generate the same amount of tax revenue to finance the spending
- A sensible government would choose the lower rate, suggested by the diagram of next page
- In equilibrium with a lower tax rate t_1, consumption, labor supply, and output are all higher, than in equilibrium with a higher tax rate t_1
- The slope of the budget constraint is $-z(1-t)$, the lower the tax rate, the steeper the budget constraint
- Thus, the lower the tax rate, the higher the utility


C
zh – G
A
I1
z(1 – t1)h
I2
F
z(1 – t2)h
H
B
h
–G

COMPARISON OF THE LUMP-SUM TAX AND THE PROPORTIONAL TAX

- With a lump-sum tax, competitive equilibrium is Pareto optimal
- A higher government spending lowers consumption and leisure
    - But output increases since labor supply is larger
- With a proportional tax, competitive equilibrium is not Pareto optimal
- A higher tax rate can raise government spending, but leads to lower consumption and more leisure
    - Output decreases since labor supply is smaller

(SKIPPED, NOT REQUIRED) 7. KEYNESIAN STICKY WAGES AND PRICES

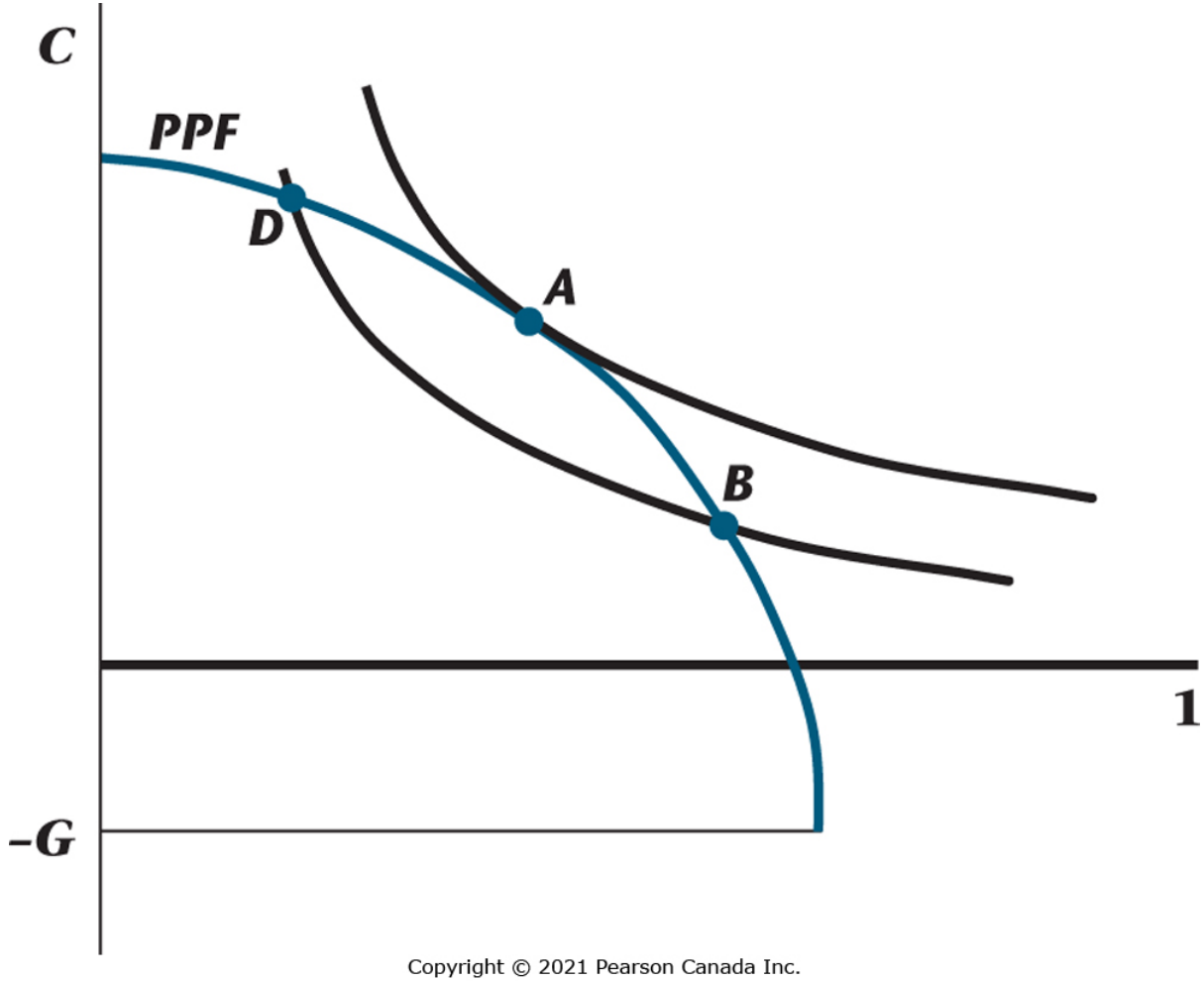
- Let us come back to to the baseline model with the lump-sum taxation
- In this model, fiscal policy (e.g. government spending) acts to increase aggregate output through an income effect on labor supply
    - A higher government spending leads to higher aggregate output, lower consumption and leisure
- There is another framework, put forward by John Maynard Keynes, in which fiscal policy (and monetary policy) works through some different mechanisms
- One main idea in Keynes' framework is call the **sticky-price theory**

- **Price stickiness** or **nominal rigidity** refers to that nominal prices (e.g., wage in terms of dollar values) change *infrequently*, due to the costs of negotiating contracts (e.g., wage contracts)
- There are empirical evidences suggesting some degree of nominal rigidity
- Some firms would rather lay off workers than retain them with lower nominal wages, evidence of *downward stickiness*
- Formal modeling price stickiness needs to incorporate money (and other nominal assets) into the model
- Here, we sketch the basic idea by using our model with the lump-sum tax, without introducing money

7-1. A SKETCH OF KEYNESIAN STICKY-WAGE MODEL

- Recall, real wage is $w = \frac{W}{P}$ where W is nominal wage and P is nominal output price
- Suppose that nominal wage W is *fixed* in the short run
    - In the long run, no price is sticky
- Also, suppose that P is *sticky*, but responsive to monetary policy and fiscal policy
    - When G increases, P increases since aggregate demand ($C + G$) is higher
    - When G decreases, P declines since aggregate demand is lower
- We now literally have a Keynesian sticky-wage model

- On the diagram in the next page, point A is the equilibrium under flexible prices, it is Pareto optimal
- When W is fixed, real wage W/P may not be socially efficient
- Point B could be an equilibrium of the Keynesian sticky-wage model. Why?
- Given real wage W/P (stickiness), the representative firm determines the labor input, by setting $MP_N = \frac{W}{P}$, and $-MP_N$ equals the slope of the PPF
- It could be that, $MP_N = \frac{W}{P}$ holds at point B, if B is an equilibrium



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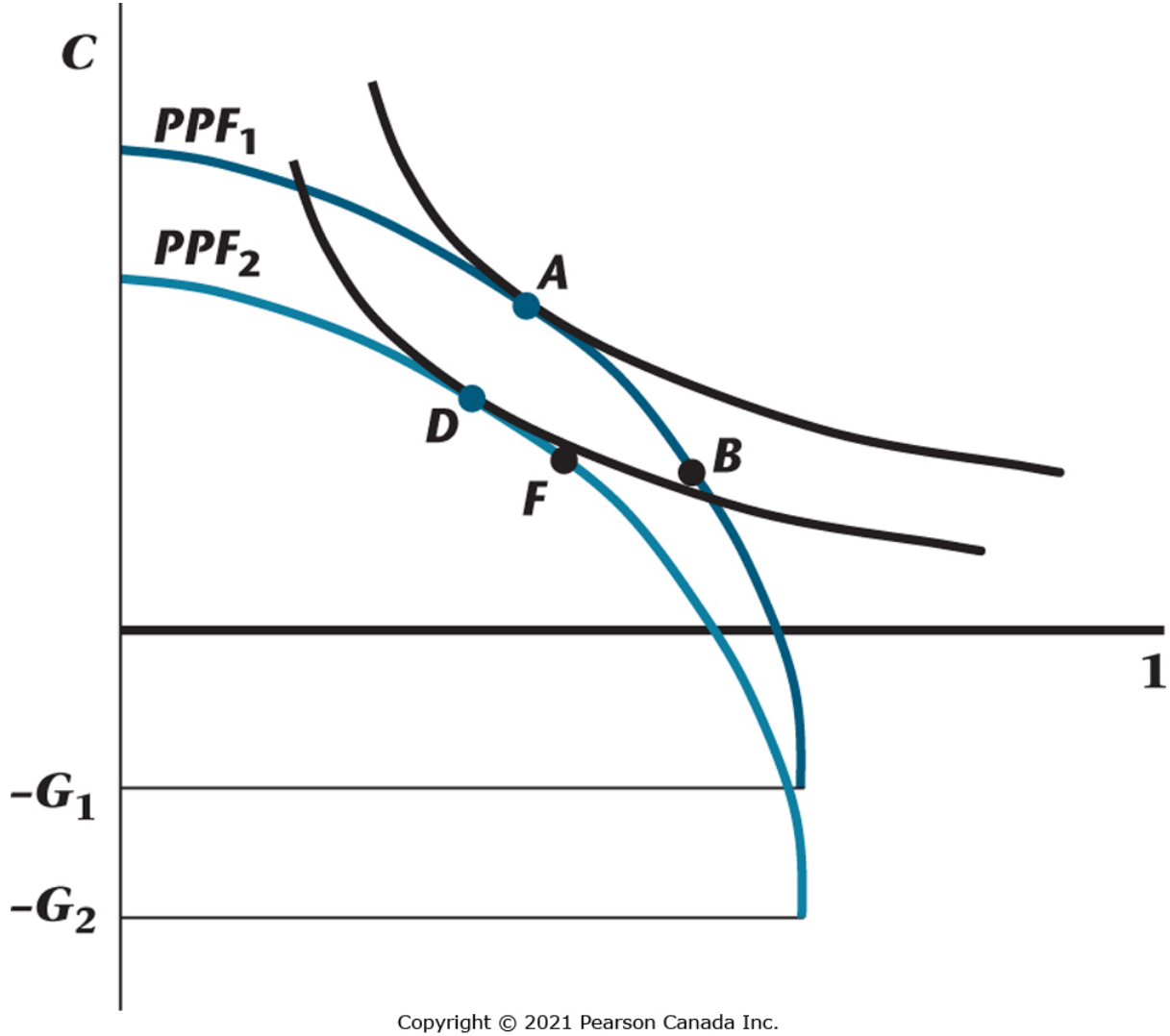
- But, at point B, the slope of indifference curve is flatter than the slope of the PPF (which in turn equals negative of real wage)
- It means that, at point B, the consumer wants to work more (and consume more of C). But, the firm chooses not to hire more labor input
- Then point B is an equilibrium in which there is *Keynesian unemployment*: excess labor supply at the market real wage

- Similarly, the equilibrium could be at point D
- At point D, the firm chooses to hire workers, but the consumer wants to enjoy more leisure given the underlying real wage
- Labor demand exceed labor supply at point D: the economy is *over-heating*
- If equilibrium in the short run is at point B or D, over time, prices will adjust. Moving to the longer run, the equilibrium will be at point A
- Thus, in the short run, there could be inefficiency due to nominal rigidity
- In the long run, the economy self-adjusts to achieve an efficient equilibrium

7-2. POLICY EFFECTS IN THE SHORT RUN

- Keynesian viewpoint is, fiscal policy and monetary policy can help the economy achieve efficiency in the short run
    - We do not need to wait for the economy adjusts itself
- Take the monetary policy, suppose that the central bank can control price P
- Real wage w at point B is inefficiently high
- If the central bank raises P to lower w, the economy can move to point A to achieve an efficient equilibrium
- Similarly, when the equilibrium is at point D, central bank can lower P to raise real wage, moving the economy to point A
- Sticky wages imply an active role for monetary policy in correcting inefficiency

- Government spending G can also help correct the short-run inefficiency of the economy
- Suppose that the initial equilibrium with sticky wages is at point B
- If government increases its spending, the PPF shifts down to PPF_2
- A higher G increases the aggregate demand, raising the price P. Real wage then falls
- It is possible that P rises sufficiently so that the equilibrium moves to point F
- Consumption is larger at point F than at point B. Not visually clear though
    - In contrast, under flexible prices, consumption is crowded out by an increase in G
- The change in aggregate output over the change of government spending, $\frac{\Delta Y}{\Delta G}$, is **government spending multiplier**



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